ETHICAL AI REVIEW PACKAGE

Patient Triage Recommendation Model

Responsible AI governance review and deployment decision

System under review	Patient Triage Recommendation Model
Deployment context	Hospital / Emergency Department — production
Governance framework	Microsoft Responsible AI Standard v2 — six principles
Dataset registry entry	DS-TRIAGE-2026-V1
Incident reference	P1-ETH-2026-004 — Priority 1
Review date	17 July 2026
Prepared by	AI Ethics & Compliance Officer
Governance decision	APPROVED WITH CONDITIONS

Executive Summary

The Patient Triage Recommendation Model was reviewed against Microsoft’s six Responsible AI principles and returned a Critical Flag for Fairness. A Priority 1 bias incident (P1-ETH-2026-004) records a Disparate Impact Ratio of 0.68 and a False Negative Rate Disparity of 0.13 against patients aged 45–60, tracing to an unaudited training distribution in dataset DS-TRIAGE-2026-V1. That dataset is rejected; both derived embedding indexes are quarantined. The model is Approved with Conditions, subject to adversarial de-biasing achieving a Disparate Impact Ratio of at least 0.80 and weekly bias audits.

Section 1 — Responsible AI Scoring

The model was scored principle by principle against Microsoft's Responsible AI Standard v2, using the Use Case Intake Form and the automated bias-scan report as the evidence base. Each rationale below references a specific characteristic of this deployment rather than a general property of AI systems.

PART A — PRINCIPLE-LEVEL SCORING

SCORING KEY

Pass — principle is adequately addressed.

Concern — potential issue requiring design changes or compensating controls.

Critical Flag — significant risk that may not be addressable through standard mitigation.

#	Principle	Score	Rationale
1	Fairness	**Critical Flag**	The bias-scan report recorded a Disparate Impact Ratio of 0.68 for patients aged 45–60, below the 0.80 fairness threshold and below the Priority 1 critical floor of 0.70.
2	Reliability and Safety	**Concern**	Systematic triage error risks delaying critical care for older patients, creating a pathway to clinical deterioration in an emergency setting where the model orders the working queue.
3	Privacy and Security	**Concern**	The intake form states a commitment to protecting patient privacy but documents no specific technical safeguards, database access controls or retention limits.
4	Inclusiveness	**Concern**	A False Negative Rate Disparity of 0.13 shows the model does not perform equitably across the adult and elderly patient groups named as its target population.
5	Transparency	**Concern**	Intake documentation contains no explainability provision and does not describe how urgency recommendations are surfaced to Emergency Room staff working at speed.
6	Accountability	**Concern**	Emergency Room staff are identified as stakeholders, but no named system owner, escalation route or human-in-the-loop audit protocol is established.

PART B — OVERALL DECISION

Decision: MITIGATE (Proceed ☐ Mitigate ☒ Reject ☐)

The tool cannot proceed directly to deployment because of a Critical Flag under Fairness, arising from documented disparate impact and false-negative disparity against the 45–60 age group. Concerns under Reliability and Safety require compensating controls to prevent clinical triage error. Release is conditionally authorised only after the mitigation retraining and audit criteria in Part C are validated.

PART C — FOLLOW-UP ACTIONS

#	Action required	Stage	Acceptance criterion
1	Implement in-processing adversarial de-biasing in the model retraining pipeline.	Pre-Release	Disparate Impact Ratio rises to ≥ 0.80 across all tested age demographics.
2	Configure automated weekly bias audits for Disparate Impact and False Negative Rate metrics.	Deployment	Weekly automated reports verified as dispatched to clinical leads.
3	Revert Emergency Department workflows to manual clinician triage while the model is paused.	Immediate	100% of triage decisions made by ER staff without AI influence during the pause.

Section 2 — Bias Findings and Mitigation Planning

2.1 Priority 1 bias findings summary

Incident reference	P1-ETH-2026-004
Impacted attribute	Age group — 45–60 relative to 25–34 reference cohort
Severity	Priority 1

Metric	Recorded	Threshold	Escalation basis
Disparate Impact Ratio	**0.68**	≥ 0.80	Below the P1 critical floor of 0.70.
False Negative Rate Disparity	**0.13**	≤ 0.05	Exceeds the maximum allowable P1 variance of 0.10.
Demographic Parity Difference	**0.12**	≤ 0.05	Exceeds the maximum allowable P1 variance of 0.10.

Three metrics breached simultaneously against the same demographic attribute. That is not measurement noise; it is a consistent directional bias in how the model assigns urgency.

WHY THIS ESCALATES TO PRIORITY 1

The False Negative Rate Disparity is the governing finding. In a triage context a false negative is not a misclassification to be corrected on the next cycle — it is a patient assigned a lower urgency band than their clinical presentation warrants, and therefore a patient who waits. A disparity of 0.13 means the model systematically under-recognises acuity in the 45–60 cohort relative to younger adults. Because the model operates in live emergency department production and its output orders the queue clinicians work through, every hour the disparity persists is measured in delayed treatment for a population whose presentations are more likely, not less, to be time-critical. The escalation is therefore severity-driven: the harm is clinical deterioration and elevated mortality risk, the exposure window is continuous, and the affected population is protected under age discrimination provisions applying to medical triage and federally funded healthcare.

CONNECTION TO THE SECTION 1 RISK

This finding is the direct evidentiary basis for the Critical Flag recorded against Fairness in Section 1. Fairness was scored a Critical Flag on the strength of the 0.68 Disparate Impact Ratio, and Inclusiveness scored a Concern on the strength of the 0.13 False Negative Rate Disparity — the same two metrics escalated here. The escalation also substantiates the Reliability and Safety Concern, since systematic triage error in a safety-critical clinical deployment is precisely the technical failure causing harm that the principle guards against. Sections 1 and 2 are not two assessments of the model; they are one finding, scored in Section 1 and escalated in Section 2.

DATA CHARACTERISTICS DRIVING THE FINDING

The disparity traces to the training distribution rather than to the model architecture. Dataset submission DS-TRIAGE-2026-V1 was accepted without an age-demographic composition analysis or a statistical bias audit — the failure recorded under Criterion 3 of the Dataset Ethics Review in Section 3. Absent representation testing at intake, the under-representation of the 45–60 cohort passed unmeasured into training, and the model learned age-correlated proxies for urgency from a sample that did not represent the population it now triages. This is why the recommended control operates on the training process rather than on the model's outputs.

Immediate action pending mitigation: automated prioritisation is paused and the Emergency Department has reverted to manual clinician triage, per follow-up action 3 of Section 1.

2.2 Mitigation options scoring

Three candidate controls were scored across the four required evaluation dimensions. Model retraining with in-processing adversarial de-biasing is the preferred option.

Evaluation dimension	Feature reweighting	Post-processing fairness correction	Model retraining (preferred)
Ethical effectiveness	**Weak.** Altering data weights does not resolve the systemic logic driving clinical de-prioritisation of older patients.	**Moderate.** Adjusts final classification outputs but leaves the biased correlations the model learned intact.	**Strong.** Adversarial de-biasing corrects the underlying mathematical bias during training itself.
Operational feasibility	**Strong.** Easy to execute on existing datasets, but does not reliably guarantee clinical safety in production.	**Strong.** Simple output-layer code adjustment with no infrastructure pipeline upgrade required.	**Moderate.** Requires new training jobs in the Azure ML pipeline, which demands compute resource.
Reversibility	**Strong.** Weights revert instantly by restoring the default pre-processing scripts.	**Strong.** The post-processing layer can be disabled immediately if output anomalies occur.	**Strong.** Retrained checkpoints are version-controlled, allowing immediate rollback to a safe baseline.
Auditability	**Moderate.** Data transform logs are auditable, but verifying structural bias reduction is complex.	**Moderate.** Tracks output shifts well but does not capture internal algorithmic change.	**Strong.** Azure ML logs all retraining configurations, fairness metrics and weights for regulatory oversight.

2.3 Preferred mitigation justification

Selected control: In-processing adversarial de-biasing with retraining

Deploying the Patient Triage Recommendation Model with an active Priority 1 bias incident introduces unacceptable clinical, ethical and legal exposure, because systematic de-prioritisation of older patients directly threatens patient safety and medical compliance. In-processing adversarial de-biasing updates the model's inner algorithmic layers so that it cannot use age-correlated proxies when calculating urgency scores, addressing the defect at its origin rather than masking it at the output.

Operationally the approach integrates with the established Azure Machine Learning pipelines and retains full version-control history, which preserves an immediate fallback to clinician-led workflows should post-deployment anomalies emerge. All fairness improvements are captured in training logs, giving the Clinical Governance Committee and the Institutional Review Board documentary evidence that the age disparity has been resolved rather than an assurance that it has.

2.4 Monitoring requirement statement

POST-IMPLEMENTATION MONITORING

> The Lead Clinical Data Scientist will implement automated batch auditing that tracks the model’s Disparate Impact Ratio, Demographic Parity Difference and False Negative Rate Disparity.
>
> Evaluations run on a strict weekly cadence against production triage data. Audit summaries are delivered to the Hospital AI Governance Committee and clinical leads to maintain continuous fairness compliance and operational safety.
>
> Any metric returning below its policy threshold re-triggers a Priority 1 escalation and re-pauses automated prioritisation.

Section 3 — Dataset Ethics Review

Dataset registry entry ID	DS-TRIAGE-2026-V1
Source	Third-party Electronic Health Record aggregator
Reviewer	AI Ethics & Data Compliance Lead
Review date	17 July 2026

3.1 Data ethics checklist

#	Criterion	Designation	Rationale
1	Sourcing Transparency	**Pass**	Vendor documentation names the EHR aggregator and maps collection methods via structured API pulls across all emergency room data segments.
2	Consent and Licensing	**Pass**	Licensing terms explicitly authorise commercial AI application, and patient consent records cover downstream clinical model training and optimisation.
3	Representational Fairness	**Fail**	No age-demographic composition or statistical bias audit was supplied, leading directly to the production Disparate Impact Ratio of 0.68 and False Negative Rate disparity of 0.13 against older patients.
4	Data Minimization	**Flag**	Dataset scope is proportionate to emergency triage needs, but the field-level data dictionary describes protected health text variables generally rather than defining pseudonymisation handling field by field.
5	Provenance Documentation	**Pass**	A complete chain of custody traces data from original emergency department intake timestamps through all aggregation and pseudonymisation cycles.

3.2 Dataset decision

Decision: REJECT (Approve ☐ Approve with conditions ☐ Reject ☒)

The third-party dataset submission for the Patient Triage Recommendation Model is rejected on a critical failure under Criterion 3, Representational Fairness. Operational bias audits established that the training distribution systematically misrepresents older patient populations, producing a False Negative Rate Disparity of 0.13 and a Disparate Impact Ratio of 0.68 against the 45–60 age group in emergency settings. Approving this data framework would create severe legal, regulatory and patient safety exposure through automated treatment de-prioritisation. The model retraining pipeline is paused until the vendor resolves these structural demographic gaps.

Criterion driving rejection: Criterion 3 — Representational Fairness.

For resubmission the vendor must supply a comprehensive demographic data audit, an explicit methodology explaining how age group representation was calculated, and a remediation plan demonstrating how the under-represented 45–60 bracket will be balanced to eliminate the operational False Negative Rate disparity.

3.3 Lineage metadata review

Upstream dependency review tracing DS-TRIAGE-2026-V1 from vendor intake through to the feature stores serving the triage model.

Source ID	Source	Status	Finding and carry-forward
SRC-01	EHR aggregator — emergency department encounter history (vitals, acuity scores, disposition, demographics) via structured API pull	**Non-compliant**	No age-demographic composition supplied and no statistical bias audit performed at intake. Criterion 3 Fail. Origin of the 0.68 and 0.13 disparities. Carries forward to IDX-TRIAGE-STRUCTURED-02 in Section 4.
SRC-02	EHR aggregator — clinical narrative and triage notes (free-text protected health information) via structured API pull	**Non-compliant**	Field-level data dictionary describes protected health text variables generally, without field-by-field pseudonymisation handling. Criterion 4 Flag. Carries forward to IDX-TRIAGE-NARRATIVE-01 in Section 4.
SRC-03	EHR aggregator — chain-of-custody and provenance metadata, intake timestamps through aggregation and pseudonymisation cycles	**Compliant**	Complete chain of custody supplied. Criterion 5 Pass. No remediation required; retained as audit evidence.

REMEDIATION PATHS

SRC-01 — demographic representation gap. The vendor must supply a full age-band composition analysis for the training distribution, the methodology used to calculate representation, and a rebalancing plan raising the 45–60 cohort to population-representative proportion. Acceptance is measured on the model rather than the document: the Disparate Impact Ratio must reach 0.80 or above across all tested age demographics, matching acceptance criterion 1 of the Section 1 follow-up actions. Until accepted, IDX-TRIAGE-STRUCTURED-02 is quarantined and excluded from retraining.

SRC-02 — pseudonymisation coverage gap. The vendor must supply a field-level data dictionary naming every protected health text variable and stating the pseudonymisation or de-identification method applied to each, with entity-level coverage evidence. Acceptance requires 100% protected health information entity coverage demonstrated on a held-out sample. Until accepted, IDX-TRIAGE-NARRATIVE-01 retains its Highly Confidential classification and its retention clock is suspended rather than running.

SRC-03. No action required. Provenance documentation is retained as supporting evidence for the governance record.

Two of three upstream sources are non-compliant, and one drives an active Priority 1 clinical bias incident. This is the substantive basis for the Reject decision above, and both non-compliant sources are carried into the Section 4 governance record.

Section 4 — Data Classification and Quality Documentation

Two embedding indexes serve the Patient Triage Recommendation Model, one built from each non-compliant upstream source identified in the Section 3 lineage review.

4.1 Classification and retention tagging record

Field	IDX-TRIAGE-NARRATIVE-01	IDX-TRIAGE-STRUCTURED-02
Index contents	Vector embeddings of clinical narrative and triage note free-text	Vector embeddings of structured encounter features — vitals, acuity scores, disposition, demographic attributes
Upstream source	SRC-02	SRC-01
Dataset registry entry	DS-TRIAGE-2026-V1	DS-TRIAGE-2026-V1
Protected health information	Yes — free-text PHI, pseudonymisation coverage unverified	Yes — indirect identifiers via demographic and encounter attributes
Sensitivity label	**Highly Confidential — PHI**	**Highly Confidential — PHI (indirect)**
Classification rationale	Criterion 4 Flag: protected health text variables lack field-level pseudonymisation handling, so re-identification risk cannot be excluded. The highest tier applies until coverage is evidenced.	Demographic and encounter attributes permit re-identification in combination. Criterion 3 Fail additionally makes this index the carrier of the age-representation defect.
Access control	Managed Identity only; no account keys or SAS tokens	Managed Identity only; no account keys or SAS tokens
Retention period	36 months from index build, aligned to the clinical model lifecycle	36 months from index build, aligned to the clinical model lifecycle
Lifecycle rule	No unhashed identifiers permitted in cold storage tiers; hard delete at expiry	No unhashed identifiers permitted in cold storage tiers; hard delete at expiry
Governance state	**Retention suspended — quarantined** pending SRC-02 pseudonymisation evidence	**Retention suspended — quarantined** pending SRC-01 demographic rebalancing
Owner	Lead Clinical Data Scientist / AI Ethics & Triage Governance Lead	Lead Clinical Data Scientist / AI Ethics & Triage Governance Lead

Because the Section 3 review rejected DS-TRIAGE-2026-V1, neither index may be promoted to a retraining or serving role. Both are held in quarantine with the retention clock suspended, preserving the evidentiary record for the vendor remediation review rather than allowing it to age out under the standard lifecycle rule. Quarantine lifts per index only when the remediation path for its upstream source is accepted.

4.2 Data quality summary

DEFINED QUALITY THRESHOLD

A feature crosses the quality threshold, and requires a remediation ticket, if field completeness falls below 95%, if protected health information pseudonymisation coverage is below 100%, or if demographic representation deviates from the served population by more than 5 percentage points.

Feature	Index	Completeness	Additional measure	Ticket
patient_age_band	STRUCTURED-02	99.2%	45–60 cohort at 11.4% of training rows against 24.8% of the served ED population — a 13.4 point deviation	**DQ-001**
chief_complaint_freetext	NARRATIVE-01	97.8%	PHI pseudonymisation coverage 62%	**DQ-002**
prior_encounter_history	STRUCTURED-02	**88.7%**	Missingness concentrated in the 45–60 cohort	**DQ-003**
vital_signs_composite	STRUCTURED-02	**93.1%**	—	**DQ-004**
triage_note_narrative	NARRATIVE-01	96.1%	PHI pseudonymisation coverage 71%	**DQ-005**
acuity_score_baseline	STRUCTURED-02	97.6%	—	Within threshold
disposition_outcome	STRUCTURED-02	96.4%	—	Within threshold

4.3 Remediation tickets

Ticket	Feature	Quality issue	Proposed remediation
DQ-001	patient_age_band	Training distribution under-represents the 45–60 cohort by 13.4 percentage points against the served ED population. Direct cause of the 0.68 Disparate Impact Ratio and 0.13 False Negative Rate Disparity.	Vendor supplies a rebalanced extract restoring age-band proportionality to within 5 points; representation re-verified before the index leaves quarantine. Closure gated on Disparate Impact Ratio ≥ 0.80, matching Section 1 acceptance criterion 1.
DQ-002	chief_complaint_freetext	PHI pseudonymisation coverage of 62% against a 100%	Vendor supplies a field-level pseudonymisation map; coverage re-

Ticket	Feature	Quality issue	Proposed remediation
		requirement — roughly two in five PHI entities in this field are unhandled.	measured on a held-out sample and evidenced at 100% before the Highly Confidential quarantine on NARRATIVE-01 is reviewed.
DQ-003	prior_encounter_history	Completeness of 88.7% against a 95% floor, with missingness concentrated in the affected age cohort — compounding the DQ-001 representation defect rather than sitting independently of it.	Vendor re-extracts prior-encounter linkage for the affected cohort; completeness re-measured to ≥ 95% with residual missingness demonstrably not age-correlated.
DQ-004	vital_signs_composite	Completeness of 93.1% against the 95% floor.	Vendor confirms whether missingness is a collection-interval artefact or an extraction defect; re-extract, or document an imputation method with clinical sign-off.
DQ-005	triage_note_narrative	PHI pseudonymisation coverage of 71% against a 100% requirement.	Covered by the DQ-002 pseudonymisation map; coverage evidenced at 100% on the same held-out sample.

4.4 Quality findings in summary

Five of seven assessed features cross the defined quality threshold, and the failures are not independent. Three of them — DQ-001, DQ-003, and the age-correlated component of DQ-004 — describe a single underlying defect: the training distribution does not represent the older adult population the model triages. That is the same defect recorded as Criterion 3 Fail in Section 3 and manifested as the Priority 1 bias incident in Section 2. The remaining two, DQ-002 and DQ-005, describe a second defect, incomplete pseudonymisation of protected health information in the free-text fields, recorded as Criterion 4 Flag in Section 3.

Both Section 3 findings are therefore present in this governance record with named features, measured values and owned remediation. Neither index may be released from quarantine until its associated tickets close.

Section 5 — Governance Decision

APPROVED WITH CONDITIONS

The decision applies to the Patient Triage Recommendation Model. It is distinct from, and conditional upon, the Reject decision recorded against vendor dataset submission DS-TRIAGE-2026-V1 in Section 3. The model has a defensible path to production; the dataset in its current form does not.

Basis across the four sections

Section 1 scored the model against Microsoft's six Responsible AI principles and returned one Critical Flag under Fairness with five Concerns, resolving to an overall decision of Mitigate rather than Reject — the defect is addressable through retraining and does not indicate an irremediable design fault. Section 2 escalated the Fairness finding to Priority 1 on the strength of three simultaneous metric breaches against the 45–60 age cohort, and selected in-processing adversarial de-biasing after scoring three candidate controls across ethical effectiveness, operational feasibility, reversibility and auditability. Section 3 traced the bias to its origin, rejecting DS-TRIAGE-2026-V1 on a Criterion 3 failure of representational fairness and identifying two non-compliant upstream sources with documented remediation paths. Section 4 classified both indexes built from those sources as Highly Confidential and quarantined, and raised five remediation tickets against the specific features carrying the defects.

The four sections describe a single causal chain: an unaudited training distribution under-represented older adults, the model learned age-correlated proxies for urgency, and the resulting triage disparity reached production. Each section addresses one link in that chain, and the recommended controls act on the origin rather than on the symptom.

Why not Approved	Why not Rejected
An active Priority 1 bias incident in a safety-critical clinical deployment cannot receive unconditional authorisation while automated prioritisation is paused and the underlying dataset stands rejected.	The defect is structural in the data, not irremediable in the system. Mitigation is feasible within existing ML pipelines, reversible through version-controlled checkpoints, and auditable to the standard the Institutional Review Board requires. Rejection would be disproportionate to a correctable representation failure with a defined remediation path.

Conditions precedent

All seven conditions must close before automated prioritisation resumes.

#	Condition	Source finding	Acceptance criterion	Owner
1	Manual clinician triage maintained; automated prioritisation remains paused.	§1 action 3 / §2	100% of triage decisions made by ER staff without AI influence during the pause.	ED Clinical Lead

#	Condition	Source finding	Acceptance criterion	Owner
2	Vendor supplies demographic audit, methodology and rebalancing plan for DS-TRIAGE-2026-V1.	§3 rejection; DQ-001	Age-band representation within 5 points of the served population.	AI Ethics & Data Compliance Lead
3	Vendor supplies field-level pseudonymisation map for protected health text variables.	§3 Criterion 4; DQ-002, DQ-005	100% PHI entity coverage evidenced on a held-out sample.	AI Ethics & Data Compliance Lead
4	Completeness remediation for prior_encounter_history and vital_signs_composite.	DQ-003, DQ-004	≥ 95% completeness, residual missingness not age-correlated.	Data Engineering
5	In-processing adversarial de-biasing applied in the retraining pipeline.	§1 action 1; §2 control	Disparate Impact Ratio ≥ 0.80 across all tested age demographics.	Lead Clinical Data Scientist
6	Quarantine lifted per index only after its upstream remediation is accepted.	§4 governance state	Both indexes cleared; sensitivity labels and retention rules re-applied.	AI Ethics & Triage Governance Lead
7	Ethical review of post-retraining metrics with Compliance, Clinical Safety and the IRB.	§2 follow-up action 3	Documented IRB sign-off.	AI Ethics & Compliance Officer

Ongoing monitoring requirement

Automated weekly batch audits run over production triage data, tracking Disparate Impact Ratio, Demographic Parity Difference and False Negative Rate Disparity. Audit summaries are delivered to the Hospital AI Governance Committee and clinical leads. The responsible owner is the Lead Clinical Data Scientist / AI Ethics & Triage Governance Lead. Any metric returning below its policy threshold re-triggers a Priority 1 escalation and re-pauses automated prioritisation. This commitment is carried unchanged from the Section 2 monitoring requirement statement and Section 1 follow-up action 2.

Lead evaluator signature	Review date
Governing officer endorsement	**Authorisation status / title**

Cross-Section Integration Check

Verification that the four sections form a single governance narrative rather than four independent artifacts.

#	Check	Result
1	Does the bias finding escalated in Section 2 connect directly to a risk identified in Section 1?	**Confirmed.** The Priority 1 escalation cites a Disparate Impact Ratio of 0.68 and a False Negative Rate Disparity of 0.13 — the same two metrics behind the Section 1 Fairness Critical Flag and Inclusiveness Concern, named explicitly in section 2.1.
2	Does the non-compliant data source flagged in Section 3 appear in the Section 4 data governance record?	**Confirmed.** SRC-01 carries to IDX-TRIAGE-STRUCTURED-02 with tickets DQ-001 and DQ-003; SRC-02 carries to IDX-TRIAGE-NARRATIVE-01 with tickets DQ-002 and DQ-005. Both the Criterion 3 Fail and the Criterion 4 Flag appear in Section 4 against named features.
3	Do all four sections consistently reference the same scenario details and data assets?	**Confirmed.** One model name, one dataset registry entry (DS-TRIAGE-2026-V1), one incident reference (P1-ETH-2026-004), one affected cohort (age 45–60 against 25–34), and one set of three metric values throughout.
4	Does the package close with a single, clearly supported governance decision?	**Confirmed.** Approved with Conditions, supported by seven numbered conditions precedent, each traced to a specific finding in Sections 1 to 4.

All four integration checks pass. No finding is introduced in a later section without an origin in an earlier one, and no finding raised in an earlier section is left without a control, an owner and an acceptance criterion.